

## ECE 332 Fall 2025 — Exam 1 Answer Key

Based on CS1 (HW1 cheat sheet) and CS2 (HW2 cheat sheet). [CS#: Section] tags show which sheet each formula comes from.

**Note on equilateral triangle area** (needed Q1):  $A = \frac{\sqrt{3}}{4}a^2$  is **not in either cheat sheet**. It is derived from the exam cover page: height of equilateral triangle  $= \frac{\sqrt{3}}{2}a$  (30-60-90 triangle, hyp  $= a$ , long leg  $= \frac{\sqrt{3}}{2}a$ ), then  $A = \frac{1}{2}bh = \frac{1}{2} \cdot a \cdot \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a^2$ . Add this formula to your next cheat sheet.

### Question 1 — Triangular Loop (20 pts)

**Setup.** Loop in  $y$ - $z$  plane,  $N$  turns, side  $a$ ,  $\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y})\sin(\omega t)$ .

[CS1 Surface Element  $d\mathbf{s}$ ]: loop in  $y$ - $z$  plane  $\Rightarrow d\mathbf{s} = \hat{x} da$ .

[CS1 Dot Products]:  $\hat{x} \cdot \hat{x} = 1$ ,  $\hat{y} \cdot \hat{x} = 0 \Rightarrow$  only the  $-2\hat{x}$  component of  $\mathbf{B}$  contributes.

$$\mathbf{B} \cdot d\mathbf{s} = B_0(-2\underbrace{\hat{x} \cdot \hat{x}}_{=1} + 3\underbrace{\hat{y} \cdot \hat{x}}_{=0})\sin(\omega t) da = -2B_0\sin(\omega t) da$$

(a) [CS1 General flux form]  $\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ .  $\mathbf{B}$  is uniform  $\Rightarrow$  [CS1 Uniform  $B$  shortcut]: multiply by area. Area (from exam cover, *not in cheat sheet*):  $A_\Delta = \frac{\sqrt{3}}{4}a^2$ .

$$\Phi = -2B_0\sin(\omega t) \cdot \frac{\sqrt{3}}{4}a^2 = -\frac{\sqrt{3}}{2}B_0a^2\sin(\omega t)$$

(b) [CS1  $N$ -turn coil]:  $V_{\text{emf}} = -N \frac{d\Phi}{dt}$ . [CS1  $\partial B/\partial t$  table]:  $\frac{d}{dt}[\sin(\omega t)] = \omega \cos(\omega t)$ .

$$V_{\text{emf}} = -N \frac{d}{dt} \left[ -\frac{\sqrt{3}}{2}B_0a^2\sin(\omega t) \right] = \boxed{\frac{\sqrt{3}}{2}NB_0\omega a^2\cos(\omega t)}$$

(c)  $N = 10$ ,  $a = 0.08$  m,  $B_0 = 0.02$  T,  $\omega = 2 \times 10^6$  rad/s,  $\omega t = \pi/4$ .

[CS1 Unit Circle]:  $\cos(\pi/4) = \frac{\sqrt{2}}{2} \approx 0.707$ . [CS1 SI Prefixes]: MHz =  $10^6$ .

$$\begin{aligned} |V_{\text{emf}}| &= \frac{\sqrt{3}}{2} \cdot 10 \cdot 0.02 \cdot (2 \times 10^6) \cdot (0.08)^2 \cdot 0.707 \\ &= 0.00554 \times 10 \times 0.02 \times 2 \times 10^6 \times 0.707 \approx 1568 \text{ V} \end{aligned}$$

$$I = \frac{V_{\text{emf}}}{R} = \frac{1568}{1000} \approx 1.57 \text{ A}$$

### Question 2 — Moving Wire (12 pts)

Wire along  $\hat{y}$ ,  $a = 0.15$  m,  $\mathbf{u} = 12\hat{x}$  cm/s  $= 0.12\hat{x}$  m/s,  $\mathbf{B} = (0.1\hat{x} - 0.05\hat{z})$  T.

[CS1 Motional EMF]:  $V_{\text{emf}} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$ .

[CS1 Cross Product (determinant)]:

$$\mathbf{u} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0.12 & 0 & 0 \\ 0.1 & 0 & -0.05 \end{vmatrix} = \hat{x}(0) - \hat{y}(0.12 \times -0.05 - 0) + \hat{z}(0) = 0.006\hat{y} \text{ V/m}$$

Wire is along  $\hat{y}$  so  $d\mathbf{l} = \hat{y} dy$ :

$$V_{\text{emf}} = \int_0^{0.15} 0.006\hat{y} \cdot \hat{y} dy = 0.006 \times 0.15$$

$$V_{\text{emf}} = 9 \times 10^{-4} \text{ V} = 0.9 \text{ mV}$$

Common mistake: forgetting to convert cm/s  $\rightarrow$  m/s (factor of 100 error).

### Question 3 — Conduction & Displacement Current (12 pts)

$I_c = 2 \sin(\omega t) \text{ } \mu\text{A}$ , cross-section  $A$ ,  $\sigma = 10^{-4} \text{ S/m}$ ,  $\epsilon_r \approx 26\pi$ .

(a) [CS1 Displacement & Conduction Current]:  $J_c = I_c/A$ , then  $J_c = \sigma E \Rightarrow E = J_c/\sigma$ :

$$E = \frac{J_c}{\sigma} = \frac{I_c/A}{\sigma} = \frac{2 \times 10^{-6} \sin(\omega t)}{10^{-4} \cdot A}$$

$$E(t) = \frac{2 \times 10^{-2}}{A} \sin(\omega t) \text{ V/m}$$

The area  $A$  is unknown and cannot be cancelled — it must stay in the answer.

(b) [CS1 Displacement & Conduction Current]:  $J_d = \epsilon_r \epsilon_0 \partial E / \partial t$ ,  $I_d = J_d \cdot A$  (the  $A$  cancels here).

[CS1  $\partial B / \partial t$  table] (same rule):  $\partial[\sin(\omega t)] / \partial t = \omega \cos(\omega t)$ .

$\omega = 10^9 \text{ rad/s}$ ,  $\omega t = \pi/3$ . [CS1 Unit Circle]:  $\cos(\pi/3) = 1/2$ .

$$\begin{aligned} I_d &= \epsilon_r \epsilon_0 \cdot \frac{2 \times 10^{-2}}{A} \cdot \omega \cos(\omega t) \cdot A = 26\pi \cdot \frac{1}{36\pi} \times 10^{-9} \cdot 2 \times 10^{-2} \cdot 10^9 \cdot \frac{1}{2} \\ &= \frac{26}{36} \times 10^{-9} \times 2 \times 10^{-2} \times 10^9 \times \frac{1}{2} = \frac{13}{18} \times 10^{-2} = \frac{13}{1800} \end{aligned}$$

$$I_d \approx 7.22 \text{ mA}$$

### Question 4 — Charge Relaxation (10 pts)

$\sigma = 10^{-4} \text{ S/m}$ ,  $\epsilon_r \approx 26\pi$ .

(a) [CS1 Charge Relaxation]:  $\tau_r = \epsilon_r \epsilon_0 / \sigma$ .

$$\tau_r = \frac{26\pi \cdot \frac{1}{36\pi} \times 10^{-9}}{10^{-4}} = \frac{\frac{26}{36} \times 10^{-9}}{10^{-4}} = \frac{13}{18} \times 10^{-5}$$

$$\tau_r \approx 7.22 \times 10^{-6} \text{ s} = 7.22 \text{ } \mu\text{s}$$

(b) [CS1 Charge Relaxation]:  $\rho_v(t) = \rho_{v0} e^{-t/\tau_r}$ . Set  $\rho_v / \rho_{v0} = e^{-3}$ :

$$e^{-t/\tau_r} = e^{-3} \Rightarrow \frac{t}{\tau_r} = 3 \Rightarrow t = 3\tau_r$$

$$t = 3 \times 7.22 \times 10^{-6} \approx 2.17 \times 10^{-5} \text{ s} = 21.7 \text{ } \mu\text{s}$$

### Question 5 — Lossless EM Wave (30 pts)

$\mathbf{E}(z, t) = 4 \cos((6\pi \times 10^8) t - kz - \pi/6) \hat{y}$  V/m.  $\varepsilon_r = 6$ ,  $\mu_r = 1$ .

(a) [CS2 Plane Wave Parameters (Lossless)]: read  $\omega$  directly from argument, then chain.

[CS2 Order:  $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$ ]

$$\omega = 6\pi \times 10^8 \text{ rad/s}$$

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{6}} \approx 1.225 \times 10^8 \text{ m/s} \quad [\text{CS2}] u_p = c/\sqrt{\varepsilon_r}$$

$$k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{6\pi \times 10^8 \cdot \sqrt{6}}{3 \times 10^8} = 2\pi\sqrt{6} \approx 15.4 \text{ rad/m} \quad [\text{CS2}] k = \omega\sqrt{\varepsilon_r}/c$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{2\pi\sqrt{6}} = \frac{1}{\sqrt{6}} \approx 0.408 \text{ m} \quad [\text{CS2}] \lambda = 2\pi/k$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{6}} \approx 153.9 \text{ } \Omega \quad [\text{CS2}] \eta = \eta_0/\sqrt{\varepsilon_r}$$

$$\omega = 6\pi \times 10^8 \text{ rad/s}, \quad k = 2\pi\sqrt{6} \text{ rad/m}, \quad \lambda = \frac{1}{\sqrt{6}} \approx 0.408 \text{ m}, \quad \eta \approx 153.9 \text{ } \Omega$$

(b) [CS1 Phasor Method]:  $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$ . Strip  $e^{j\omega t}$ , keep the rest:

$$\tilde{\mathbf{E}}(z) = 4e^{-jkz} e^{-j\pi/6} \hat{y}$$

$$\tilde{\mathbf{E}}(z) = 4e^{-j(kz+\pi/6)} \hat{y} \text{ V/m}$$

(c) [CS2 Direction of Propagation table]: argument contains  $-kz \Rightarrow \hat{k} = +\hat{z}$ .

[CS2  $\hat{k} \times \hat{E}$  table]:  $\hat{k} = +\hat{z}$ ,  $\hat{E} = \hat{y} \Rightarrow \hat{H} = -\hat{x}$ .

[CS2 Lossless Plane Wave  $\mathbf{H}$  from  $\mathbf{E}$ ]:  $\tilde{\mathbf{H}} = \frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$ .

$$\hat{k} \times \tilde{\mathbf{E}} = \hat{z} \times (4e^{-j(kz+\pi/6)} \hat{y}) = 4e^{-j(kz+\pi/6)} (\hat{z} \times \hat{y}) = -4e^{-j(kz+\pi/6)} \hat{x}$$

$$\tilde{\mathbf{H}}(z) = \frac{-4}{\eta} e^{-j(kz+\pi/6)} \hat{x} = -\frac{\sqrt{6}}{30\pi} e^{-j(kz+\pi/6)} \hat{x} \text{ A/m} \approx -0.026 e^{-j(kz+\pi/6)} \hat{x} \text{ A/m}$$

(d) [CS2 Phasor  $\leftrightarrow$  Time Domain table]: real phasor coeff  $A \rightarrow A \cos(\omega t - \beta z)$ .

$$\mathbf{H}(z, t) = -\frac{4}{\eta} \cos(6\pi \times 10^8 t - 2\pi\sqrt{6} z - \frac{\pi}{6}) \hat{x} \approx -0.026 \cos(6\pi \times 10^8 t - 2\pi\sqrt{6} z - \frac{\pi}{6}) \hat{x} \text{ A/m}$$

(e) [CS2 Average Power Density (Poynting) lossless shortcut]:  $\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$ .

$|E_0| = 4 \text{ V/m}$ ,  $\hat{k} = +\hat{z}$ :

$$\mathbf{S}_{\text{av}} = \frac{4^2}{2 \times 153.9} \hat{z} = \frac{16}{307.8} \hat{z}$$

$$\boxed{\mathbf{S}_{\text{av}} \approx 0.052 \hat{z} \text{ W/m}^2}$$

### Question 6 — EM Wave in Distilled Water (16 pts)

$$f = \frac{2}{\pi} \text{ GHz}, \sigma = 10^{-4} \text{ S/m}, \varepsilon_r = 26\pi, \mu_r = 1.$$

$$[\text{CS2 Plane Wave Parameters}]: \omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}.$$

$$(a) [\text{CS2 Lossy Media — Classification, loss tangent}]: \frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega \varepsilon_r \varepsilon_0}.$$

$$\varepsilon_r \varepsilon_0 = 26\pi \cdot \frac{1}{36\pi} \times 10^{-9} = \frac{13}{18} \times 10^{-9} \text{ F/m}.$$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{10^{-4}}{4 \times 10^9 \times \frac{13}{18} \times 10^{-9}} = \frac{10^{-4}}{\frac{52}{18}} = \frac{10^{-4} \times 18}{52} = \frac{18}{52} \times 10^{-4}$$

$$\boxed{\frac{\varepsilon''}{\varepsilon'} = \frac{9}{26} \times 10^{-4} \approx 3.46 \times 10^{-5}}$$

(b) [CS2 Lossy Media — Classification table]:

| $\varepsilon''/\varepsilon'$ | Type                                                              |
|------------------------------|-------------------------------------------------------------------|
| $> 100$                      | Good conductor                                                    |
| $0.01$ to $100$              | Quasi-conductor                                                   |
| $< 0.01$                     | Low-loss dielectric ← <b>here:</b> $3.46 \times 10^{-5} \ll 0.01$ |

Distilled water at this frequency is a **low-loss dielectric**.

$$(c) \text{ Since low-loss dielectric, use [CS2 Low-Loss Medium] approximation: } \alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma \eta}{2}.$$

First find  $\eta$  ([CS2 Lossless params], valid for low-loss):

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{26\pi}} \approx \frac{376.99}{9.04} \approx 41.7 \Omega$$

$$\alpha \approx \frac{\sigma \eta}{2} = \frac{10^{-4} \times 41.7}{2} \approx 2.09 \times 10^{-3} \text{ Np/m}$$

$$[\text{CS2 Lossy Media (bottom strip)}]: \delta_s = 1/\alpha:$$

$$\boxed{\delta_s = \frac{1}{\alpha} \approx \frac{1}{2.09 \times 10^{-3}} \approx 479 \text{ m}}$$

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**Score summary (if perfect):** Q1: 20/20, Q2: 12/12, Q3: 12/12, Q4: 10/10, Q5: 30/30, Q6: 16/16 = **100/100**